



DesignWare SDRAM and Static Memory Controller Installation Guide

DW_memctl

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Synopsys, Inc.
700 E. Middlefield Road
Mountain View, CA 94043

www.synopsys.com

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Preface

About This Manual

This manual describes how to download and install the DesignWare SDRAM and Static Memory Controller (DW_memctl), as well as related tools and libraries. It also describes how to set up your environment, licenses, and tools. This manual is primarily intended for system administrators, but also contains information of interest to anyone using the DW_memctl.

Install Installation Guide Organization

The chapters in this guide are organized as follows:

- Chapter 1, “[Installing the DW_memctl Component](#)” provides step-by-step instructions of how to download this release image of the DW_memctl.
- Chapter 2, “[Installation Instructions for Supported Tools](#)” provides step-by-step instructions of how to download supported versions of Verification IP, tools and operating systems for this DW_memctl release.
- Chapter 3, “[Supported Versions for DW_memctl and Tools](#)” lists supported versions of Verification IP, tools, and operating systems for this DW_memctl release.
- Chapter 4, “[Setting Up Your Environment](#)” describes the environment variables that must be set, and specifies license requirements for components.
- Chapter A, “[Troubleshooting Download/Installation Issues](#)” describes several methods you can use to check your components and tools for Synopsys updates.
- Chapter B, “[Updating a DesignWare Library](#)” discusses how to update your DesignWare library.

Related Documents

To see a complete listing of documentation available for the DW_memctl, refer to the [Guide to Documentation for DesignWare SDRAM and Static Memory Controller](#).

Web Resources

- DesignWare IP product information: <http://www.designware.com>
- Your custom DesignWare IP page: <http://www.mydesignware.com>
- Documentation through SolvNet: <http://solvnet.synopsys.com> (Synopsys password required)
- Synopsys Common Licensing (SCL): <http://www.synopsys.com/keys>

Customer Support

To obtain support for your product:

- First, prepare the following debug information, if applicable:
 - For environment setup problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, use the following menu entry:
File > Build Debug Tar-file

Check all the boxes in the dialog box that apply to your issue. This menu entry gathers all the Synopsys product data needed to begin debugging an issue and writes it to the file `<core tool startup directory>/debug.tar.gz`.
 - For simulation issues outside of coreConsultant or coreAssembler:
 - Create a waveforms file (such as VPD or VCD)
 - Identify the hierarchy path to the DesignWare instance
 - Identify the timestamp of any signals or locations in the waveforms that are not understood
- Then, contact Support Center, with a description of your question and supplying the above information, using one of the following methods:
 - *For fastest response*, use the SolvNet website. If you fill in your information as explained below, your issue is automatically routed to a support engineer who is experienced with your product. The **Sub Product** entry is critical for correct routing.
Go to <http://solvnet.synopsys.com/EnterACall> and click on the link to enter a call. Provide the requested information, including:
 - **Product:** DesignWare Library IP
 - **Sub Product:** Memory
 - **Tool Version:** *<product version number>*
 - **Problem Type:**
 - **Priority:**
 - **Title:** DW_memctl
 - **Description:** For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood

After creating the case, attach any debug files you created in the previous step.

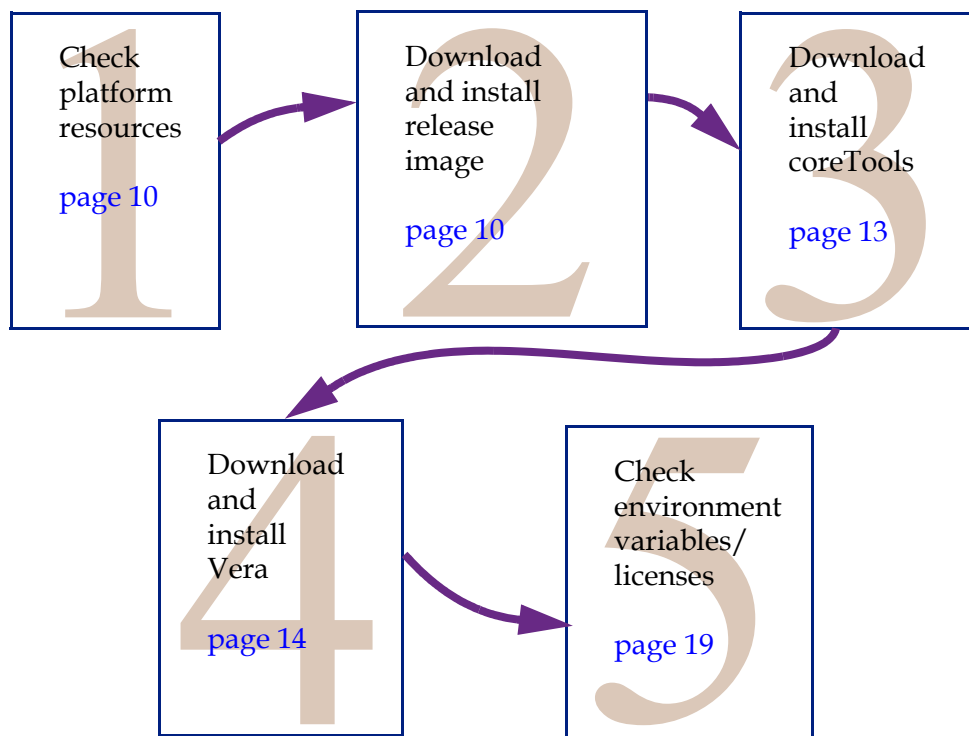
- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product name, Sub Product name, and Tool Version number in your e-mail (as identified above) so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created in the previous step.
- Or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<http://www.synopsys.com/Support/GlobalSupportCenters>

Installing the DW_memctl Component

This chapter describes how to download and install the DW_memctl. The download image *does not* contain related tools, so you must download those separately; this chapter directs you to information for downloading related tools.

1.1 Overview of Download and Installation Process

To use the DW_memctl, you must perform the following setup steps.



For an overview of all product documentation for the DW_memctl, refer to the [Guide to Documentation for DesignWare SDRAM and Static Memory Controller](#).

1.2 Recommended Platform Resources

The following are recommended resources for the platform to which you are downloading and installing software mentioned in this document.

- The DW_memctl release image installation (which includes VIP used for simulation of the DesignWare synthesizable components) requires 64MB of disk space to install.
- The coreTools image installation requires an additional 433 MB of disk space.
- The Vera image can require up to 455 MB of disk space, depending on which version you require.
- Your platform must be running an operating system listed on [18](#).
- All downloads require anonymous FTP access to <ftp.synopsys.com>, or authenticated access through the Release Library.

1.3 Installing DW_memctl

You can download the .run file in two ways:

- **Static Image** – Sends a static image containing all product components.
- **Dynamic** – Analyzes a DESIGNWARE_HOME tree and sends an image containing only those components not already installed or in need of update. In other words, the image from an optimized download is customized to contain only the components that a specific DESIGNWARE_HOME tree requires in order to be brought up-to-date with respect to the standard image. The net effect on your DESIGNWARE_HOME after installation is the same as if you had installed the standard image.

“[Installation Instructions for Supported Tools](#)” on page [13](#) describes how to download and install the required Synopsys tools as well. Refer to “[Supported Versions of Tools and Libraries](#)” on page [17](#) for supported versions of these tools; not all versions are supported.

1.3.1 Static Image

The following outlines the installation instructions for the static release image.

Perform the following steps:

1. To download the latest DW_memctl image, go to the following link – also called the DWDL page:
https://www.synopsys.com/dw/dwdl.php?id=iip_dw_memctl
2. Once you have downloaded the .run file, follow these installation instructions:
 - a. Change permissions on the downloaded .run file:

```
% chmod u+x image_name.run
```
 - b. Create a DESIGNWARE_HOME environment variable to point at your install location, which must be read/write mounted to all compute servers. All users must set this environment variable. For C-shell:

```
% setenv DESIGNWARE_HOME full_UNIX_path/DESIGNWARE_HOME
```
 - c. Execute the .run file to unpack the image:

```
% image_name.run $DESIGNWARE_HOME
```

If you have the five source license package for DW_memctl—DWC-MEMCTL-Source—enter it and provide the project ID associated with your source license when prompted during installation.

- d. You must ensure that the \$DESIGNWARE_HOME/vip directory has both read and write permissions for anyone using the verification models for simulation.
- e. Modify your \$PATH variable to include \$DESIGNWARE_HOME/bin.

1.3.2 Dynamic

The following outlines the installation instructions for the dynamic release image.

Perform the following steps:

1. To download the latest DW_memctl image, go to the following link—also called the DWDL page:
https://www.synopsys.com/dw/dwdl.php?id=iip_dw_memctl
2. Follow the instructions for executing the dwh_update.run utility.

Installation Instructions for Supported Tools

This chapter describes how to download and install the supported tools and libraries for the DW_memctl. The download image of a particular release *does not* contain related tools, examples and libraries, so you must download those separately.

2.1 coreConsultant and coreAssembler (coreTools)

The coreTools are required to configure DesignWare synthesizable IP components. The coreAssembler tool is required in order to configure multiple DesignWare IIP in a single subsystem. The coreConsultant tool configures only a single DesignWare IIP component, but you can also use coreAssembler for one component. For more information, please refer to the following page:

http://www.synopsys.com/products/designware/ipreuse_tools.html

The coreTools are not included in the DesignWare IP component download image; follow instructions on this page to download and install the coreTools.

2.1.1 Downloading coreTools

If you or your site administrator have not yet downloaded the release image, this is your first step.

1. Go to <https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>.
Enter your SolvNet ID and Password when prompted.
2. Choose the product name (coreAssembler, coreTools, or coreConsultant – all three point to the same set of coreTools download files) from the “My Product Releases” selection list.
3. Choose the appropriate version from the “Version” selection list. For more information about supported versions for this DW_memctl release, refer to “[Supported Versions of Tools and Libraries](#)” on page 17.
4. Click the **Download Here** button to download from the Synopsys EST server.
5. Click on “Yes, I AGREE TO THE ABOVE TERMS” button to accept the Synopsys download terms.
6. Select the appropriate platform-specific download image.

2.1.2 Installing coreTools

To install coreTools, use this procedure:

1. Issue the following commands (assumes tar.gz files are in the root of the install tree):


```
% cd download_dir
% gtar zxvf coreTools_version.common.tar.gz
% gtar zxvf coreTools_version.platform.tar.gz
```
2. Add the path for *download_dir/bin* to your path variable definition. For example:


```
% set path=( $path download_dir/bin )
```
3. Set the LM_LICENSE_FILE to a server that has a coreAssembler or coreConsultant license.
4. Ensure that coreConsultant and coreAssembler are properly installed by invoking each of them.


```
% coreConsultant &
% coreAssembler &
```

If either tool fails to invoke, do not perform any additional steps until you have resolved the problem by checking all of the previous steps in this procedure.

2.2 DesignWare Verification IP

The DW_memctl is included in this release image, so there is no need to download it separately. For additional information about DesignWare Verification IP, refer to the *Guide to the DesignWare SDRAM and Static Memory Controller* at:

https://www.synopsys.com/dw/doc.php/vip/dw_memctl/latest/doc/intro.pdf

2.3 Vera

In the testbenches provided with DesignWare Library IP, the DesignWare Verification IP requires Vera for simulation. Note, however, that a Vera license is not required.

To download and install Vera:

1. Go to the SolvNet Download Center page:

<https://solvnet.synopsys.com/DownloadCenter/dc/product.jsp>
2. Choose Vera from the “My Product Releases” selection list.
3. Choose the appropriate version from the version selection list. For more information about supported versions for this release of the DW_memctl, refer to “[Supported Versions of Tools and Libraries](#)” on page 17.
4. Click the **Download Here** button to download from the Synopsys EST server.

You may want to view the Release Notes before installing. If so, select “Click here to view release notes” link.

5. Click on “Yes, I AGREE TO THE ABOVE TERMS” button to accept the Synopsys download terms.
6. Download the vera_INSTALL_README.txt file and follow the directions to complete the installation.

7. Set the \$VERA_HOME environment variable to point to your Vera install location.
8. Modify your \$PATH variable to include \$VERA_HOME/bin and \$VERA_HOME/lib.
9. Ensure that Vera is properly installed by issuing the following commands:

```
% echo $PATH  
% echo ${VERA_HOME}
```

The first command should list \$VERA_HOME/bin and \$VERA_HOME/lib among the search locations. The second command should return the path to your Vera install location

2.4 DesignWare Building Blocks for DC

For information on how to install the supported versions of DesignWare Building Blocks for Design Compiler, refer to the “Installation” section of the *DesignWare Building Block IP DC Release Notes* at:

https://www.synopsys.com/dw/doc.php/doc/dwf/manuals/dwf_dc_rn.pdf



A supported version of DesignWare Building Blocks may already exist on your system if Design Compiler or Physical Compiler have already been installed. See the contents of the \$SYNOPSYS/dw/version file to determine what version of DesignWare Building Blocks is currently installed.

2.5 Keeping your Components and Tools Current

Synopsys provides a command called dwh_update that is included in your DESIGNWARE_HOME. This command compares the versions of your components, tools, and technologies with the latest that are available from Synopsys. It generates a report only; it does not automatically make updates. For more information on using this command, see “[Updating a DesignWare Library](#)” on page 23.

3

Supported Versions for DW_memctl and Tools

This chapter lists supported versions of related tools for this release of the DW_memctl.

3.1 Supported Versions of Tools and Libraries

The following table lists the supported versions of tools and libraries used by this release of the DW_memctl. Specific versions that have been tested and are known good at the time of this release are listed in the table, but newer versions of each tool are supported as well.

Table 3-1

Tool	Version
Design Compiler (DC) – current	2010.12-SP3
Tetramax	2010.12-SP3
Design Compiler (DC) – oldest allowable	2009.06-SP5
Tetramax	2009.06-SP5
coreTools	2010.09-SP2; download information on 13
Formality	2010.12-SP3
PrimeTime	2010.12-SP3
Leda	2010.03
Vera	2009.12-6
VCS	2010.06-SP1
Synplify	2011.03

3.2 Supported Versions of Verification Model Suites and VMT

The following table lists the supported versions of verification models and simulators used in conjunction with this release of the DW_memctl.

Table 3-2

Tool	Supported Versions
DesignWare VIP for AMBA model suite	6.00b
DesignWare VMT	3.70a
Vera on Linux Red Hat 4.0 or Solaris 9	2009.12-6
	Download information is on 14 .

3.3 Supported Operating Systems

The follow operating systems are supported with the tools and IP listed in this installation guide. A simulator-specific operating system support table is provided in “[Supported Simulators and Operating Systems](#)” on page 18.

- Solaris 9 (32-bit)
- Red Hat Enterprise Linux 4.0 (32-bit, AMD64)¹

Current release updates should be maintained on these operating systems.

3.4 Supported Simulators and Operating Systems

The following table shows the Vera, VMT, and operating system versions supported for each simulator.

Table 3-3

Simulator	for Vera 2009.12-6/VMT 3.70a
Synopsys VCS/VCSi 2010.06-SP1 *	Solaris 9/64 ** Red Hat Enterprise Linux 4.0/32
MTI Verilog 6.6b for DW_axi_a2x; 6.5b for all others	Solaris 9/64 ** Red Hat Enterprise Linux 4.0/32
Cadence NC-Verilog 9.20-s030	Solaris 9/64 ** Red Hat Enterprise Linux 4.0/32
* DW_memctl does not support VCSi	
** DW_memctl does not support 64-bit operating systems.	

1. DW_memctl does not support 64-bit operating systems.

4

Setting Up Your Environment

This section describes the environment variables and licenses you need in order to use the DW_memctl.

4.1 Tool Installation Roots

You should check with your system administrator to determine if all tool installation roots are set so that coreAssembler and coreConsultant can find the necessary synthesis and verification tools. You can set tool installation roots from within coreAssembler and coreConsultant using the **Edit > Set Tool Installation Roots** pull-down menu.

Simulation tool roots are picked up from environment variables every time coreAssembler and coreConsultant are invoked and are not set by **Set Tool Installation Roots**.

4.2 Environment Variables

[Table 4-1](#) describes the environment variables required for the DW_memctl.

Table 4-1 Environment Variables for DW_memctl

Environment Variable	Description
DESIGNWARE_HOME	Path to installed DesignWare synthesizable and verification components.
LM_LICENSE_FILE	Provides access to tool and IP licenses. Alternatively, SNPSLMD_LICENSE_FILE may be used.
LD_LIBRARY_PATH	Path to operating system libraries (Solaris and Linux). Required if you run coreAssembler or coreConsultant simulations.
SYNOPSYS	Path to the Synopsys synthesis tools tree. Required if you do not have a source license or if you want to run coreAssembler or coreConsultant synthesis.
VERA_HOME	The path to where you installed Vera.
GCC_HOME	Path to GCC install directory for Solaris and Linux.

Table 4-1 Environment Variables for DW_memctl (Continued)

Environment Variable	Description
PATH	Must include the following paths: \$DESIGNWARE_HOME/bin Path to coreTools “bin” directory If you run coreAssembler or coreConsultant simulations, must also include: \$VERA_HOME/bin \$VERA_HOME/lib Paths to simulator binaries and libraries (see specific simulator in this table). If you do not have a source license or run coreAssembler or coreConsultant synthesis, must also include: \$SYNOPSYS/platform/syn/bin
VCS Users (NOTE: define VCS_HOME -or- VCSi_HOME, depending on the VCS simulator you use)	
VCS_HOME	Path to VCS install directory. \$PATH must include \$VCS_HOME/bin and \$VCS_HOME/lib.
VCSi_HOME	Path to VCSi install directory. \$PATH must include \$VCSi_HOME/bin and \$VCSi_HOME/lib.
NC-Verilog Users	
PATH	Include the path for <i>path_to_NC-Verilog/tools/bin</i> .
ModelSim Verilog or MTI	
PATH	Include the path for <i>MTI_install_dir/modeltech/bin</i> .

4.2.1 Licenses

The DesignWare SDRAM and Static Memory Controller require a DesignWare license to create configured, encrypted RTL for verification or synthesis using coreAssembler or coreConsultant. In order for you to create unencrypted RTL source, you must have the source license for the specific component, as shown in [Table 4-2](#).

In order to make full use of the DesignWare synthesizable components, you need additional licenses for the Synopsys and third-party tools for synthesis, TetraMax, Formality, PrimeTime, and HDL simulation. A Vera license is not required to run the verification environment supplied with the component.

Table 4-2 License Requirements for DW_memctl

Component Name	Encrypted RTL License	Unencrypted RTL Source Licenses	Included VIP License with Unencrypted RTL Source License
DW_memctl	DesignWare	DWC-MEMCTL-Source	AMBA VIP and DWMM Suites
coreAssembler Using DesignWare Library IP	DesignWare or coreAssembler		

A

Troubleshooting Download/Installation Issues

Q. How can I make sure that the source installation was done properly?

A. You can make sure that the source installation was done properly in one of two ways:

- Go to the `$DESIGNWARE_HOME/iip/component/latest/auxiliary` directory and check to see that the `.component` file is present in this directory. For example, if you have a source license for the DW_ahb, the `.DW_ahb` file should be present in the “auxiliary” directory.
- Configure the IP in cC or cA and check the `workspace/src` directory for source RTL.

Q. I have a DesignWare license and installed the image properly. When I invoke cC or cA, I get the following message. Why?

Command not found

A. You get this message because coreTools was not installed. It is not a part of an image and has to be installed separately.

B

Updating a DesignWare Library

B.1 Are Your Components and Tools Current?

Synopsys provides several methods to let you know if your components and/or tools are current or out of date:

- You can subscribe to MyDesignware notifications on a component basis, where you receive an email when a component updates or has new STAR information; refer to “[myDesignware Subscriptions](#)”.
- You can use the command `dwh_update` (in `DESIGNWARE_HOME/bin`) to help you determine the versioned objects (components, tools) in your `DESIGNWARE_HOME`, and to help manage updates to these entities; refer to “[dwh_update Command](#)”.
- You can enable automatic update checking in `coreAssembler` and `coreConsultant`, which checks the components in your design against both your DesignWare Library, and the currently supported Synopsys components; refer to “[coreTools Automatic Update Checking](#)”.

B.1.1 myDesignware Subscriptions

- myDesignWare enables you to receive product information that is of interest to you, such as product updates, technical articles, in-depth application notes and much more. You can add or remove selected subscriptions at any time. Sign-up through your SolvNet user account at:

<https://www.synopsys.com/dw/mydesignware.php>

- DesignWare Technical Bulletin Subscriptions

Subscribe to a quarterly publication for Synopsys DesignWare customers, which contains technical information regarding DesignWare products such as in-depth application notes.

- DesignWare Component Subscriptions

Proactive notification of new releases, STAR information availability and more. To add a component to your subscription list, locate it using the Search for IP tool on the left and select the “Subscribe” link.

B.1.2 dwh_update Command

The `dwh_update` command (`$DESIGNWARE_HOME/bin` directory) performs the following functions:

- Downloads and installs updates to images you may have installed previously. The script optimizes the download process such that it fetches only the portions of an image which are newer than what is already installed.
- Provides a quick way to list the components and their versions in your `DESIGNWARE_HOME` tree.

B.1.2.1 dwh_update Usage

To display usage instructions for the current version of `dwh_update`, execute the `-help` switch:

```
% dwh_update -help
```

When `dwh_update` is executed, it automatically checks to see if it is the latest `dwh_update` available. If it is a non-current version, it will update itself (automatically download and install the latest version).

B.1.2.2 Installation Reports

Synopsys' customer support representatives may ask you to run a `dwh_update -info` report on your `DESIGNWARE_HOME` tree in order to debug an installation problem. The following command string is a typical example invocation:

```
% dwh_update -info 1 -r /remote/project_1/reports
```

This command string writes a list of the latest components installed in your `DESIGNWARE_HOME` tree to a file named `dwh_latest.rpt`, which will reside in the directory `/remote/project_1/reports`.

B.1.3 coreTools Automatic Update Checking

`coreAssembler` and `coreConsultant` provide a scheduled way for you to check for DesignWare Library synthesizable and verification component updates as well as existing STARs for the Synopsys IP used in your design workspace. When you complete the Add Subsystem Component activity in `coreAssembler` or the Specify Configuration activity in `coreConsultant`, these tools check your component version(s) against the most recent version(s) available *both* for download from Synopsys, and in your local `DESIGNWARE_HOME` library. A report gives you newer version information, if available, and lists STARs created/fixed for the component(s) you are using.

You can perform a manual check at any time using the menu item **Help > Check for IP Updates**, or set an interval for `coreTools` to automatically generate the report.



Attention

Components are not automatically updated; this operation only generates a report. You must make these component updates manually.

For more information about Automatic/Manual IP update checks in `coreAssembler` and `coreConsultant`, see:

- "Component Update Checking" in the [coreAssembler User Guide](#)
- "Component Update Checking" in the [coreConsultant User Guide](#)